

Studies from
databases/registers (**n = 905**)

Science Direct (n = 423)

Scopus (n = 239)

Web of Science (n = 196)

IEEE Xplore (n = 47)

References from other
sources (**n =**)

Citation searching (n =)

Grey literature (n =)

References removed (**n = 178**)

Duplicates identified manually (n = 0)

Duplicates identified by Covidence (n = 178)

Marked as ineligible by automation tools
(n = 0)

Other reasons (n =)

Studies screened (**n = 727**)

Studies excluded (**n = 623**)

Studies sought for retrieval (**n = 104**)

Studies not retrieved (**n = 0**)

Studies assessed for eligibility (**n = 104**)

Studies excluded (**n = 48**)

Full text not available (n = 3)

Wrong outcome — no performance metrics
reported (n = 11)

Wrong intervention — no deep learning
method used (n = 11)

Wrong population — not students or e-
learning context (n = 22)

Not a full study — editorial, opinion, or
book chapter (n = 1)

Studies included in review (**n = 56**)